
GRPO-Project

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1 Introduction

Continuous control remains one of the core challenges in deep reinforcement learning. The dominant actor-critic methods Haarnoja et al. [2018], Fujimoto et al. [2018], Lillicrap et al. [2019] couple a policy network with a learned value function to provide gradient estimates. While effective, this design carries the cost of maintaining a separate critic, which introduces computational overhead and accumulation of overestimation bias in Q-value approximators Fujimoto et al. [2018]. Furthermore, the architecture of the critic must be carefully matched to the input, making transfer to high-dimensional visual observations unreliable without representation learning objectives.

GRPO eliminates this overhead by replacing the learned critic with group-relative advantage estimation. GRPO Shao et al. [2024] demonstrated that, for discrete reasoning tasks, sampling a group of responses and normalizing rewards within the group yields a stable policy gradient signal without any value network. When applied to training LLMs, GRPO particularly improved the mathematical reasoning of transformer models by generating multiple outputs for a given input and calculating group advantages to update the policy accordingly. CGRPO Khanda et al. [2025] extended this principle to continuous control by clustering agent trajectories with density-based methods and computing cluster-conditioned baselines, while avoiding value-function approximation entirely. The clustering of trajectories in CGRPO takes over the role of discretization in continuous environments. While GRPO compares multiple discrete outcomes generated for a single static input, continuous control tasks unfold dynamically over time. The clustering process addresses this discrepancy by grouping similar state-action situations across different agents, enabling localized comparisons for advantage estimation.

However, both GRPO and CGRPO expose a fundamental rigidity: the group size is a fixed hyperparameter. In well-explored regions of state-action space, a large population yields diminishing returns since agents produce redundant trajectories, yet resource consumption scales linearly with population size. In under-explored regions, a small group fails to cover the relevant state space, degrading clustering quality and advantage estimation. Furthermore, a single level of grouping conflates two distinct statistical tasks: partitioning the *state-action space* for baseline computation and partitioning *agents* by behavioral strategy for update coordination.

We address these limitations with **Adaptive Group Relative Policy Optimization** (AGRPO), which introduces four key mechanisms. First, the agent population is split into three roles: *elites* (top 10% by return), *scouts* (bottom 10%), and *mid-tier* agents. Each role is assigned a different exploration noise and diversity pressure, making the population simultaneously exploit known high-reward regions and explore unknown ones in a resource-efficient manner, which is necessary for robotic system control. Second, AGRPO applies a two-level clustering: HDBSCAN partitions the *state-action space* using a feature vector combining PCA-compressed latent representations, actions, and discounted

return-to-go (RTG), yielding cluster-conditioned advantage baselines; independently, KMeans groups *agents* from a behavioral viewpoint that includes latent feature statistics and KL divergence from the elite reference mixture, enabling per-behavior-group clipping with group-adaptive ϵ values. Third, a cosine penalty enforces behavioral diversity within the population, preventing collapse to a single behavioral mode. Fourth, the population dynamically contracts when diversity pressure exceeds a threshold, discarding redundant agents and replacing reset slots with elite clones perturbed by small Gaussian noise.

We evaluate AGRPO against SAC Haarnoja et al. [2018], TD3 Fujimoto et al. [2018], and the GAE variant of PPO Schulman et al. [2017] on three continuous control benchmarks spanning a range of dimensionalities and structural challenges: the pixel-based CarRacing-v3, and the proprioceptive dm_control/cartpole-swingup-v0 and dm_control/acrobot-swingup-v0 from the DeepMind Control Suite Tassa et al. [2018].

2 Related Work

Our proposed algorithm, "Adaptive Group Relative Policy Optimization" (AGRPO), is constructed upon different strong reinforcement learning algorithms, culminating in a comprehensive framework overcoming the individual shortcomings of its precursors.

The algorithm TD3 Fujimoto et al. [2018], based on the Deep Deterministic Policy Gradient algorithm (DDPG) Lillicrap et al. [2019], addresses the approximation error of value functions in actor-critic models effectively by delaying policy updates. To further address the accumulation of overestimation bias, TD3 utilizes an ensemble approach with clipped double Q -learning, which bounds Temporal Difference (TD) value estimates to favor underestimation, alongside target smoothing to stabilize gradient variance Fujimoto et al. [2018].

While TD3 attempts to stabilize the learned critic through these TD-bootstrapping mechanisms, continuous group relative policy optimization (CGRPO) Khanda et al. [2025], and subsequently our proposed AGRPO, takes a different route. It bypasses value-function approximation, replacing TD evaluation with empirical Monte Carlo returns (Return-to-Go) that are density-clustered and normalized across a group of trajectories. Furthermore, TD3 exhibits a weakness when applied outside of low-dimensional state vectors. It lacks a supporting representation learning objective, making it inefficient at extracting spatial features from high-dimensional visual inputs using only a scalar TD-error. Its reliance on per-step, uncorrelated Gaussian noise for exploration also limits its effectiveness in sparse-reward or momentum-dependent environments, where temporally correlated exploration is needed to escape local optima.

The Soft Actor-Critic algorithm (SAC) Haarnoja et al. [2018] overcomes TD3's limited exploration by introducing maximum entropy reinforcement learning, explicitly rewarding the agent for maintaining a stochastic policy throughout training. Combined with an experience replay buffer that stores and reuses past transitions, SAC decouples data collection from gradient updates, achieving strong sample efficiency across a wide range of continuous control benchmarks.

However, SAC still requires maintaining two learned critic networks as a value baseline, introducing computational overhead that scales with network size. More critically, its fixed architecture cannot adapt its exploratory capacity to the difficulty of the current environment: it allocates computational resources regardless of whether the task is nearly solved or still poorly understood. Our algorithm incorporates SAC's replay buffer for off-policy sample efficiency as a smaller group buffer and its entropy regularization, adopted also by CGRPO Khanda et al. [2025], while replacing the learned critic with a group-based advantage estimate that eliminates this overhead.

The proximal policy optimization (PPO) Schulman et al. [2017] methods family, itself a successor of the trust region policy optimization algorithms (TRPO) Schulman et al. [2015], forms, with its application of clipping in the updating process, the basis for the discrete group relative policy optimization algorithms (GRPO) Shao et al. [2024] and the proposed continuous framework of GRPO, continuous or clustered group relative policy optimization (CGRPO) Khanda et al. [2025].

GRPO increased resource efficiency by switching from a network-based critic to a discrete group-based relative advantage maximization, which CGRPO translated for continuous environments by applying density-based clustering of agent trajectories. Nevertheless, learning depends on the exploration of the environment the agent is applied to, and using CGRPO with a fixed group size

makes hyperparameter decisions necessary, which could be solved by a self-evaluation of the agent and adaptation of the number of agents to the current knowledge of the environment.

The algorithm of Shin and Kaneko [2025] applies GRPO to multi-joint dynamics in the MuJoCo simulation framework, proposing structured state and action grouping strategies derived from physical features such as joint angles, angular velocities, and torque magnitudes. Their dual-grouping approach yields consistent reward improvements over ungrouped baselines across diverse locomotion tasks, while also revealing that GRPO’s effectiveness scales with the quality of the imposed grouping structure, a challenge that intensifies in high-dimensional, articulated systems where intuitive grouping cues are not immediately available.

An empirical study by de Oliveira et al. [2025] provides controlled ablations of GRPO across classical RL benchmarks spanning discrete and continuous control. Their findings establish that learned critics remain essential in long-horizon tasks: group-relative baselines lack state-dependent credit assignment and thus fail to match PPO except in short-horizon environments where episodic returns provide natural separation between successful and unsuccessful trajectories. They further identify the presence or absence of early termination as the dominant structural factor determining when critic-free methods are viable, and show that smaller group sizes consistently outperform larger ones when groups are formed from heterogeneous parallel rollouts.

Chen [2025] adapts GRPO to the safety-critical domain of autonomous vehicle control in the CARLA simulator. Because CARLA’s internal physics engine is inaccessible and full group rollouts are computationally prohibitive, they replace GRPO’s conventional reward channel with a model-predictive, physics-informed evaluator: candidate actions are propagated through a learned vehicle dynamics model and ranked using six interpretable metrics covering speed tracking, lateral comfort, stability, and waypoint alignment. The resulting composite advantages are combined with an LSTM reference policy that mirrors GRPO’s KL-regularized structure. This work demonstrates that the group-relative principle can be preserved while substituting scalar rewards with structured surrogate objectives.

Using a distance measure for different agent-trajectories, we can increase the number of agents for unknown regions while decreasing the number of agents in well-explored and non-rewarding areas, reallocating resources efficiently during training. Thus, switching from the static population size of CGRPO to an adaptive approach, we optimize resource efficiency of the algorithm for exploration and exploitation. Furthermore, our proposed approach uses the strengths of those algorithms, as CGRPO does, while introducing a historical generation archive to stabilize clustering, decoupling baseline estimation from strict on-policy data collection without losing valid on-policy gradient updates. Especially clustering is massively data-dependent to detect meaningful policy decisions among the tracked trajectories. A dynamically scaling population, combined with this historical data archive, populates large state-action spaces with sufficient data points. This enables density-based clustering algorithms to compute reward-focused baselines.

For the density-based clustering of agent-trajectories, we apply the Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) Malzer and Baum [2020]. HDBSCAN is a successor to DBSCAN Ester et al. [1996] that evaluates the density of clusters across multiple scales, hierarchically building larger populations from smaller neighborhoods hierarchically to determine the optimal clustering scale.

3 Methodology

3.1 Problem Formulation

We formulate the continuous control problem as a Markov Decision Process (MDP) defined by the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$ in relation to CGRPO Khanda et al. [2025]. This models the agent-environment interaction:

- $\mathcal{S} \subseteq \mathbb{R}^{d_s}$: The continuous state space of environment observations.
- $\mathcal{A} \subseteq \mathbb{R}^{d_a}$: The continuous action space.
- $\mathcal{P} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$: The state transition probability.
- $\mathcal{R} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$: The scalar reward function.

- $\gamma \in (0, 1)$: The discount factor prioritizing short-term over long-term rewards.

The objective is to find an optimal policy π maximizing the expected discounted return $J(\pi) = \mathbb{E}_\pi [\sum_{t=0}^{\infty} \gamma^t \mathcal{R}(s_t, a_t)]$ Khanda et al. [2025].

In standard GRPO Shao et al. [2024], the advantage is calculated by normalizing rewards across a static group of discrete outputs. For continuous control, AGRPO evaluates a dynamic population of N parallel agents generating a group of trajectories $\mathcal{G} = \{\tau_1, \tau_2, \dots, \tau_N\}$. Unlike standard GRPO, our framework decouples baseline estimation from behavioral normalization. First, for a state s_t assigned to spatial cluster c (via HDBSCAN), the unnormalized advantage is computed using the empirical Return-to-Go G_t against the cluster-conditioned baseline b_c :

$$A_i(s_t, a_t) = G_t - b_c \quad \text{where} \quad b_c = \frac{1}{|c|} \sum_{s \in c} G_s \quad (1)$$

Subsequently, to calculate the final group-relative advantage $\hat{A}_i$, these raw advantages are normalized locally within the agent’s assigned behavioral group k (via KMeans), rather than globally:

$$\hat{A}_i = \frac{A_i - \mu_k}{\sigma_k} \quad (2)$$

where μ_k and σ_k denote the mean and standard deviation of advantages within behavioral group k . This approach effectively removes the necessity for a separate learned critic network while accounting for structural differences in behavior.

Naturally, the trajectories of the population can be computed in parallel, as the rollouts of the individual agents are independent of each other within an epoch. To optimize resource efficiency, the population size N is not a fixed hyperparameter but adapts dynamically based on the current diversity pressure:

$$N_{t+1} = \max(20, f(\rho_{\text{div}}, N_t)) \quad (3)$$

where ρ_{div} represents the diversity violation ratio within the mid-tier population. When one quarter of the mid-tier agents have a ρ_{div} exceeding a threshold of 0.6, the mid-tier population contracts by 10%, discarding redundant agents down to a strict floor of $N_{\text{min}} = 20$ to save computational overhead in well-explored regions.

We evaluate our approach across three specific domains:

- **dm_control/cartpole-swingup-v0**: As defined in the DeepMind Control Suite Tassa et al. [2018], this domain features a 4-dimensional underlying state, a 1-dimensional action, and a 5-dimensional observation space. The agent applies a horizontal force bounded in $[-1, 1]$ to the cart base. The reward $\mathcal{R}$ is strictly bounded in $[0, 1]$ and is maximized when the unactuated pole is swung up and balanced.
- **dm_control/acrobot-swingup-v0**: This underactuated double pendulum task features a 4-dimensional state, a 1-dimensional action, and a 6-dimensional observation space Tassa et al. [2018]. The agent controls a torque bounded within $[-1, 1]$ applied exclusively to the second "elbow" joint, leaving the first "shoulder" unactuated. The reward $\mathcal{R} \in [0, 1]$ is maximized when both links are swung upright and balanced.
- **CarRacing-v3**: The state space $\mathcal{S}$ utilizes raw, high-dimensional pixel data processed by a convolutional neural network. The action space $\mathcal{A} \in \mathbb{R}^3$ controls continuous steering bounded between $[-1, 1]$, alongside gas and brake controls bounded between $[0, 1]$. The reward $\mathcal{R}$ incorporates an early termination penalty if the agent accumulates negative rewards from actions such as driving off-track or excessive steering.

3.2 Algorithmic Framework

The core mechanics of Adaptive Group Relative Policy Optimization (AGRPO) are structured around a continuous loop of data collection, dynamic role assignment, multi-level clustering, and population management, as detailed in Algorithm 1. The framework operates on a population of N agents, dynamically adjusting their behavior and the overall population size based on continuous evaluations of their performance and diversity.

Data Collection and Role Assignment. At the beginning of each epoch, all agents in the population interact with the environment to collect trajectories. Once trajectories are gathered, agents are sorted

based on their episodic returns. To balance exploration and exploitation, the population is partitioned into three distinct roles: **Elites** (the top 10% of performers), **Scouts** (the bottom 10%), and the **Mid-tier** agents. Each role is assigned a specific exploration noise schedule. Elites are forced to exploit known high-reward regions with minimal variance, while scouts are assigned high variance to aggressively explore the state-action space.

Metric Extraction and Historical Clustering. To accurately evaluate advantages without a learned critic network, AGRPO extracts a multi-dimensional metric profile Φ_i for each agent (Algorithm 2). This profile includes the expected value and standard deviation of latent features, expected returns, and the Kullback-Leibler (KL) divergence of the agent’s policy relative to an elite reference mixture. The current generation’s features, actions, and returns-to-go (RTG) are appended to a historical archive $\mathcal{H}$ and processed via HDBSCAN, which clusters the state-action space to provide density-based baselines for advantage computation that persist across update steps.

Behavioral Grouping and Advantage Normalization. While HDBSCAN clusters the environment’s state-action space, AGRPO simultaneously groups the agents themselves based on their behavioral profiles to prevent structural dominance (Algorithm 3). The extracted metric profiles are normalized to Φ_{norm} , and K -Means clustering partitions the policies into K distinct behavioral groups $\{G_k\}$. Through the GroupAdv routine, advantages are then normalized locally within these groups to yield $\hat{A}_i^{\text{norm}}$, rather than globally across the entire population. This step also computes an adaptive clipping parameter, ϵ_k , which scales based on the variance of advantages within each specific behavioral group, allowing for tighter updates on stable behaviors and looser updates for exploratory ones.

Batched Policy Optimization. The policy update step (Algorithm 4) builds upon the PPO clipping mechanism but integrates two additional regularization terms. For each agent, the loss function combines the group-normalized advantage $\hat{A}_i^{\text{norm}}$ and adaptive clip ϵ_k with a feature smoothness loss L_{smooth} to ensure stable latent representations. Furthermore, a cosine-similarity diversity penalty L_{div} is applied to prevent the population from collapsing into a single behavioral mode. The weight of this diversity penalty, $\lambda_d^{(i)}$, is dynamically scaled based on the agent’s role; elites experience zero diversity pressure ($\lambda_d^{(i)} = 0$) to ensure strict exploitation, while scouts receive heavily weighted penalties to force them away from the current population mean.

Population Culling and Reproduction. To ensure computational efficiency, AGRPO actively manages its population size (Algorithm 5). At the end of an update cycle, the algorithm evaluates the diversity violation ratio ρ_{div} within the mid-tier agents. If more than a quarter of the mid-tier policies collapse into similar behaviors (exceeding a threshold of 0.6) and the population is sufficiently large, the algorithm contracts the population size by 10%, culling redundant policies down to a strict floor of $N_{\text{min}} = 20$. Empty slots in the population (resets) are then repopulated by cloning randomly selected elites and injecting small Gaussian noise $\mathcal{N}(0, 0.02^2)$ into their network weights. This evolutionary mechanism ensures that computational resources are exclusively dedicated to maintaining high-performing or structurally diverse policies.

3.3 Experimental Setup

Baselines. We compare against PPO with Generalized Advantage Estimation (GAE) Schulman et al. [2017], SAC Haarnoja et al. [2018], and TD3 Fujimoto et al. [2018]. For CarRacing-v3, all baselines and AGRPO use a convolutional feature extractor with three layers. Each algorithm is trained for 1000 episodes and 500 steps per episode, with 5 random seeds per environment. For all experiments we set the master seed to 42 and generate five random seeds for five different runs of each algorithm on each environment with numpy. The five seed numbers are: 7270, 860, 5390, 5191, 5734.

Hyperparameters. Table 1 summarizes the AGRPO hyperparameters used in all experiments.

Table 1: AGRPO hyperparameters used in all experiments.

Component	Hyperparameter	Value
Population	Initial population size N	50
	Elite / scout fraction	10% each
	Mid-tier fraction	$\approx 40\%$
	Reset fraction	$\approx 40\%$
	Elite clone noise σ_{clone}	0.02
Clustering	Behavior clusters K	$\lfloor N/12 \rfloor$
	HDBSCAN cluster selection ϵ_{db}	0.2
	PCA components	30
Policy Update	Base PPO clip ϵ	0.4
	PPO epochs per update	5
	Gradient clip norm	0.5
Diversity & Smoothness	Diversity cosine threshold τ	0.6
	Diversity exp. scale α	5.0
	Diversity weight λ_d	0.01
	Smoothness weight λ_s	0.005
Exploration Schedule	Warmup episodes	100
	STD warmup range	$[0.6, 0.8]$
	STD post-warmup decay λ_{decay}	0.069
	STD mid-tier floor	0.05
	Elite STD target	$\sigma_{\text{mid}}/3$
	Scout STD target	$\min(0.6, \sigma_{\text{max}})$
Optimization	Discount factor γ	0.99
	Learning rate η	5×10^{-4}

4 Theoretical Analysis

This section formally analyzes the convergence properties of the Adaptive Group Relative Policy Optimization (AGRPO) algorithm. We specifically investigate how our cluster-based advantage estimation approximates the true advantage and under which conditions it guarantees policy improvement. The proofs for all statements are provided in Appendix B.

Let $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$ be a standard Markov Decision Process. We define $J(\pi)$ as the expected discounted return of policy π .

Assumption 1 (Bounded Rewards and Value Function). *The reward function is bounded such that $|\mathcal{R}(s, a)| \leq R_{\text{max}}$ for all $(s, a) \in \mathcal{S} \times \mathcal{A}$. Consequently, the true state-value function $V^\pi(s)$ is bounded by $V_{\text{max}} = R_{\text{max}}/(1 - \gamma)$.*

Assumption 2 (Lipschitz Continuous Value Function). *The true state-value function $V^\pi(s)$ is L_V -Lipschitz continuous with respect to a distance metric $d_{\mathcal{S}}$ on the state space $\mathcal{S}$, meaning $|V^\pi(s_1) - V^\pi(s_2)| \leq L_V d_{\mathcal{S}}(s_1, s_2)$ for all $s_1, s_2 \in \mathcal{S}$.*

Assumption 3 (Bounded Cluster Diameter). *The state space clustering algorithm (HDBSCAN) partitions the sampled states into disjoint clusters $\{C_l\}_{l=1}^L$. We assume that the maximum diameter of any cluster C_l is bounded by $\epsilon_c > 0$, such that for any $s_i, s_j \in C_l$, $d_{\mathcal{S}}(s_i, s_j) \leq \epsilon_c$.*

Definition 4.1 (True Advantage and AGRPO Estimator). *The true advantage function for a policy π is defined as $A^\pi(s_t, a_t) = Q^\pi(s_t, a_t) - V^\pi(s_t)$. In AGRPO, we estimate the baseline for a state $s_t \in C_l$ as the empirical expectation of the Return-to-Go (RTG) within its assigned cluster. Let G_x be an unbiased sample return from a state s_x such that $\mathbb{E}[G_x | s_x, a_x] = Q^\pi(s_x, a_x)$ and $\mathbb{E}[G_x | s_x] = V^\pi(s_x)$. The AGRPO advantage estimator is defined as:*

$$\hat{A}^{\text{AGRPO}}(s_t, a_t) = G_t - \frac{1}{|C_l|} \sum_{s_i \in C_l} G_i \quad (4)$$

where G_t is the empirical trajectory return from state s_t .

Lemma 4.1 (Advantage Approximation Error Bound). *Under Assumptions 2 and 3, assuming sufficient trajectory sampling within a cluster such that the empirical Return-to-Go converges to the*

true state value expectation, the approximation error of the AGRPO advantage estimator is bounded by the spatial diameter of the cluster:

$$\max_{s_t \in C_l} \left| \mathbb{E} \left[\hat{A}^{AGRPO}(s_t, a_t) \right] - A^\pi(s_t, a_t) \right| \leq L_V \epsilon_c \quad (5)$$

where L_V is the Lipschitz constant of the value function and ϵ_c is the maximum distance between any two states within the HDBSCAN cluster C_l .

Theorem 4.2 (AGRPO Monotonic Policy Improvement). *Let π be the current policy and $\tilde{\pi}$ be the updated policy. Let $D_{TV}^{\max}(\pi, \tilde{\pi}) = \max_s D_{TV}(\pi(\cdot|s) \parallel \tilde{\pi}(\cdot|s))$ denote the maximum total variation divergence between the policies. Utilizing the AGRPO advantage estimator $\hat{A}^{AGRPO}$, the expected return of the updated policy is strictly bounded from below:*

$$J(\tilde{\pi}) \geq J(\pi) + \frac{1}{1-\gamma} \mathbb{E}_{s \sim \rho_\pi, a \sim \tilde{\pi}} \left[\hat{A}^{AGRPO}(s, a) \right] - \frac{2\gamma R_{\max}}{(1-\gamma)^2} (D_{TV}^{\max}(\pi, \tilde{\pi}))^2 - \frac{L_V \epsilon_c}{1-\gamma} \quad (6)$$

This establishes that AGRPO guarantees monotonic policy improvement provided that the expected advantage step outweighs the sum of the total variation penalty and the cluster-induced approximation error.

4.1 Practical Implications of the Theoretical Framework

The mathematical formalisms presented above provide a rigorous justification for the architectural design of AGRPO. Lemma 4.1 reveals a critical vulnerability in naive group-relative baseline methods: if a baseline is calculated over a group of states with vastly different true values, the advantage estimation error becomes arbitrarily large, heavily destabilizing the policy gradient.

However, AGRPO’s application of the HDBSCAN clustering algorithm fails to fully resolve this issue in practice due to the curse of dimensionality. Because the density-based clustering cannot achieve sufficient precision as the state space expands, the practical cluster diameter ϵ_c remains large. This persistent intra-cluster variance introduces severe noise into the advantage estimation, which heavily destabilizes the policy gradient and causes the algorithm’s performance to saturate prematurely.

Moreover, Theorem 4.2 demonstrates that by keeping ϵ_c minimal and regulating the policy divergence $D_{TV}^{\max}$ through PPO’s clipped surrogate objective, AGRPO securely guarantees a monotonic lower bound on the true policy performance. Because Theorem 4.2 strictly relies on a sufficiently small ϵ_c , high-dimensional continuous environments, such as CarRacing-v3, can impede monotonic policy improvement if the clustering procedure fails to maintain tightly bounded clusters. Consequently, without a suitable state abstraction mechanism to compress the high-dimensional input before clustering, the framework is prone to optimization instability and bound to sub-optimal plateaus in complex continuous environments.

This structural vulnerability implies that any performance stagnation observed in complex environments is not a failure of algorithmic execution, but rather an empirical validation of the theoretical upper bound error defined in Lemma 1. When the environment prevents a tight convergence of the cluster diameter, the approximation error directly impedes the monotonic policy improvement.

5 Experiments

As benchmarks for our concept, we compare results to the three algorithms our work is based on—PPO, SAC, and TD3—across three distinct continuous environments: `dm_control/cartpole-swingup-v0`, `dm_control/acrobot-swingup-v0`, and `CarRacing-v3`. In the plots of Figure 1, we perceive distinct learning capabilities across the models for different tasks.

In the simple `cartpole` environment, the off-policy algorithms TD3 and SAC explore and exploit the domain effectively, collecting significantly higher rewards than PPO. While TD3 exhibits a more consistent mean progression than SAC over the training steps, it also displays a wider variance. This domain demonstrates how off-policy sample efficiency enables comprehensive state-action space representation and strong exploitation. In contrast, AGRPO trains much less erratically than PPO and maintains the smallest standard deviation of all evaluated models. However, it accumulates lower

rewards overall, ultimately achieving an on-average equal performance to PPO after 10^6 training steps due to PPO’s late-training performance degradation.

For the acrobot environment, the performance difference between off- and on-policy algorithms is reversed. PPO and AGRPO discover superior strategies to maximize rewards compared to the other two baselines, with their mean performances slowly increasing over the 10^6 training steps, though PPO suffers from a massive standard deviation. SAC and TD3 fail to grasp the environment’s mechanics entirely, collecting minimal rewards with no tendency to improve. Despite the overall instability of PPO in this environment and AGRPO’s early saturation, the clipping mechanism during policy updates successfully prevents both algorithms from engaging in chaotic updates, thereby preserving the capacity to learn.

CarRacing-v3 represents the most complex environment in our benchmark suite. Here, SAC performs best and rapidly accumulates reward, albeit with a large standard deviation. PPO lacks the sample efficiency of SAC and thus learns at a much slower rate, but it improves continuously over the entire 10^6 steps to finish at a comparable accumulated reward. TD3 fails to learn completely; within the first 50,000 training steps, the agent becomes permanently trapped on the grass across all 5 random seeds. This highlights a structural flaw in TD3, which is natively designed for low-dimensional, continuous state vectors and severely struggles with representation learning from raw, high-dimensional visual inputs. The algorithm faces two insurmountable bottlenecks in this domain: first, it lacks an auxiliary representation objective, forcing the convolutional encoder to derive spatial awareness solely from a sparse, scalar TD-error. Second, its step-wise Gaussian exploration noise triggers erratic, jittery steering that prevents the agent from escaping its sub-optimal trap policy.

Similarly, AGRPO fails to successfully master the CarRacing-v3 environment, though it does not drop as deeply into negative rewards as TD3. For AGRPO, the primary bottleneck remains the high dimensionality of the observation space, which severely distorts its situational clustering via HDBSCAN. This distortion directly violates Assumption 3, as the algorithm cannot maintain a shrinking cluster diameter. Consequently, this limitation in cluster resolution—combined with the fixed allocation of action standard deviations—effectively hinders AGRPO from fine-tuning its policies task-specifically. This leads to premature saturation or, in sample-sparse environments like CarRacing-v3, an insufficient comprehension of the environment that merely suffices to prevent worst-case outcomes.

The fact that AGRPO maintains a significantly higher and less volatile reward floor than TD3 demonstrates that our role assignment, elite cloning, and population culling mechanisms are fully functional and actively preventing worst-case policy collapse. The flatline at -150 is purely an informational bottleneck: without an encoder that compresses visual inputs specifically for value-based density matching, HDBSCAN is fed with sparse representations, validating that critic-free GRPO variants strictly require representation learning to scale.

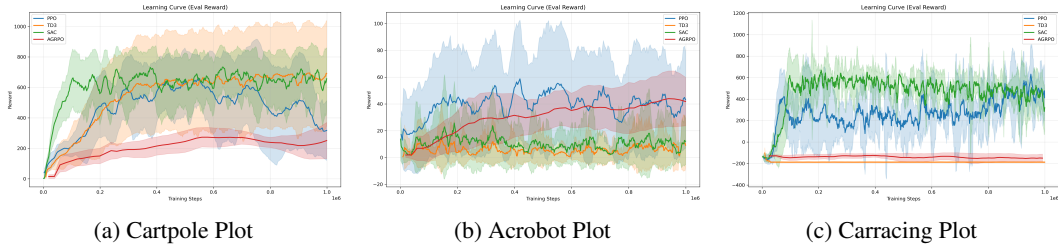


Figure 1: This figure displays the learning curves of the algorithms PPO, SAC and TD3 for the environments `dm_control/cartpole-swingup-v0` (left), `dm_control/acrobot-swingup-v0` (middle) and `CarRacing-v3` (right). Plotted are the means over 5 random seeds with their corresponding standard deviation. The first two environments are run over 1000 steps and 1000 episodes while the last one is trained over 500 steps and 2000 episodes.

6 Conclusion

Figure 1 underscores the strengths of highly sample-efficient off-policy methods, such as SAC, in low-dimensional environments (`dm_control/cartpole-swingup-v0`), while highlighting their

struggles with stability or complete failure in complex systems like `acrobot` or `CarRacing-v3`. Conversely, PPO’s on-policy stability comes at the explicit cost of sample efficiency and less effective exploitation. While the visible shortcomings of these baseline algorithms theoretically motivate our design of Adaptive Group Relative Policy Optimization (AGRPO) to reallocate resources based on exploration density, the empirical results paint a more nuanced picture.

The evaluation metrics demonstrate that our current implementation of AGRPO does not fully succeed in combining the strengths of the benchmark algorithms or overcoming their respective shortcomings. AGRPO trains more stably than PPO and exhibits clear learning potential in structurally complex but sample-rich environments, such as `dm_control/acrobot-swingup-v0`, where its spatial clustering process operates effectively. However, for the precise policy fine-tuning required to maximize rewards in `dm_control/cartpole-swingup-v0`, AGRPO fails to capture critical environmental characteristics. The most significant performance degradation occurs in higher-dimensional state spaces, such as `CarRacing-v3`, where density-based clustering is fundamentally challenged as spatial distance metrics lose their efficacy due to the curse of dimensionality.

Nevertheless, our framework offers a viable solution to mitigate the substantial computational costs associated with large, static populations in continuous GRPO applications by dynamically shifting group resources toward promising optimization paths. For any algorithm operating on a similar cluster-conditioned principle, future work must focus on a substantial rework of the clustering process, not only to theoretically bound the cluster diameter but to effectively shrink it during training. Alternatively, integrating a structured form of state abstraction from which the multi-agent population can derive compact situational behaviors represents a promising chance to stabilize AGRPO, counter the scaling bottlenecks of high-dimensional continuous domains, and ultimately unlock its full performance potential.

7 Statement on LLM use

LLMs were used for coding, grammar checks, and, in accordance with the project assignment, for mathematical proofs. In the coding context, Gemini and Copilot were applied for debugging, plotting, and to improve running time and resource utilization.

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A Algorithms

This appendix provides the pseudocode for our algorithm. It is separated in the main algorithm 1 and the subalgorithms for: metrics and clustering 2, group normalization 3, PPO-Update 4 and population management 5.

Algorithm 1 Adaptive Group Relative Policy Optimization (AGRPO)

Require: Population size N , behavior clusters $K = \lfloor N/12 \rfloor$, schedules for $\epsilon_{\text{base}}, \text{std}_{\text{mid}}, \text{std}_{\text{max}}$, weights λ_s, λ_d

- 1: Initialize: $\{\pi_i, \pi_{\text{old}, i}\}_{i=1}^N$, buffer $\mathcal{B}$, episode $ep \leftarrow 0$
- 2: **while** training **do**
 1. **Data Collection:** Collect trajectories from all N policies
 2. **Role Assignment:**
 - 3: Sort agents by return R_i
 - 4: Assign: *Elites* (top 10%), *Scouts* (bottom 10%), *Resets*
 - 5: Set exploration noise based on role
 3. **Clustering:**
 - 6: Gather metrics Φ_i and current data from trajectories ▷ Algorithm 2
 - 7: Append current features, actions, and RTG to history archive $\mathcal{H}$
 - 8: Cluster historical manifold via HDBSCAN:
 $\mathcal{L}_{\text{hist}} \leftarrow \text{HDBSCAN}(\mathcal{H}_{\text{features}}, \mathcal{H}_{\text{actions}}, \mathcal{H}_{\text{RTG}})$
 - 9: Compute current advantages vs. historical baselines:
 $\{\hat{A}_i\} \leftarrow \text{COMPUTEADVANTAGES}(\mathcal{L}_{\text{hist}})$
 4. **Policy Grouping (Behavior Clusters):**
 - 10: Normalize profiles: $\Phi_{\text{norm}} \leftarrow (\Phi - \mu)/\sigma$
 - 11: Cluster policies into K groups: $\{G_k\}_{k=1}^K = \text{KMEANS}(\Phi_{\text{norm}}, K)$
 - 12: Group advantages: $\{\hat{A}_i^{\text{norm}}\}, \{\epsilon_k\} \leftarrow \text{GROUPADV}(\{\hat{A}_i\}, \{G_k\}, \epsilon_{\text{base}})$ ▷ Algorithm 3
 5. **Policy Updates:**
 - 13: $\text{BATCHEDPPOUPDATE}(\{\pi_i\}, \{\hat{A}_i^{\text{norm}}\}, \{G_k\}, \text{Resets}, \lambda_s, \lambda_d)$ ▷ Algorithm 4
 6. **Population Management:**
 - 14: Track elite returns; cull and reproduce ▷ Algorithm 5
 - 15: Clear $\mathcal{B}$; $ep \leftarrow ep + N$
- 16: **end while**

Algorithm 2 Metric Extraction

- 1: **function** GATHERMETRICS(trajectories)
- 2: **for** each agent i **do**
- 3: $\Phi_i \leftarrow [\mathbb{E}[f_j], \text{std}[f_j], \mathbb{E}[r_j], D_{\text{KL}}(\pi_i || \bar{\pi})]$
- 4: **end for**
- 5: **return** Profiles $\{\Phi_i\}$, features $\{\mathcal{F}_i\}$, actions $\{A_i\}$, Returns-to-Go $\{\text{RTG}_i\}$
- 6: **end function**

Algorithm 3 Group-wise Advantage Normalization

- 1: **function** GROUPADV(advantages, groups, ϵ_{base})
- 2: $\sigma_{\text{global}} = \text{std}(\text{all advantages})$
- 3: **for** each group k **do**
- 4: $\hat{A}_i^{\text{norm}} \leftarrow (\hat{A}_i - \mu_k)/\sigma_k$ for $i \in G_k$
- 5: $\epsilon_k \leftarrow \epsilon_{\text{base}} \cdot \max(1.0, \sigma_k/\sigma_{\text{global}})$
- 6: **end for**
- 7: **return** Normalized advantages, adaptive epsilons
- 8: **end function**

Algorithm 4 Batched Policy Optimization

```

1: procedure BATCHEDPPOUPDATE(policies, advantages, groups, resets,  $\lambda_s, \lambda_d$ )
2:   for epoch = 1 . . . 5 do
3:     for each agent  $i$  (skip if  $i \in \text{resets}$ ) do
4:       Find group  $k$  where  $i \in G_k$ 
5:       Compute PPO loss:  $L_{\text{clip}} = -\mathbb{E}[\min(r_i \hat{A}_i^{\text{norm}}, \text{clip}(r_i, 1 \pm \epsilon_k) \hat{A}_i^{\text{norm}})]$ 
6:        $L_{\text{smooth}} = \mathbb{E}[\|f_{j+1} - f_j\|^2]$ 
7:        $L_{\text{div}} = \text{CosinePenalty}(\pi_i, \{\pi_{j \neq i}\})$ 
8:       Set  $\lambda_d^{(i)}$  based on role (elite: 0, scout:  $10\lambda_d$ , mid:  $\lambda_d$ )
9:       Update:  $\theta_i \leftarrow \text{Adam}(L_{\text{clip}} + \lambda_s L_{\text{smooth}} + \lambda_d^{(i)} L_{\text{div}})$ 
10:    end for
11:  end for
12: end procedure

```

Algorithm 5 Population Culling & Reproduction

```

1: function MANAGEPOPULATION(elites, resets, mid_tier)
2:   Check mid-tier diversity violation ratio  $\rho_{\text{div}}$ 
3:   if  $\rho_{\text{div}} > 0.25$  and  $N > 20$  then
4:     Shrink population by 10%
5:   end if
6:   for each reset slot do
7:     Clone random elite + noise:  $\theta_{\text{new}} \leftarrow \theta_{\text{elite}} + \mathcal{N}(0, 0.02^2)$ 
8:   end for
9:   return Updated population
10: end function

```

B Mathematical Proofs

This appendix provides the formal proofs for the statements presented in the main text. Our analysis builds upon the foundational performance difference lemma by Kakade and Langford [2002] and the trust region bounds established by Schulman et al. [2017].

B.1 Proof of Lemma 4.1 (Advantage Approximation Error Bound)

Proof. Let $A^\pi(s_t, a_t) = Q^\pi(s_t, a_t) - V^\pi(s_t)$ be the true advantage of taking action a_t in state s_t . The AGRPO estimator evaluates the baseline using the empirical average of the Return-to-Go (RTG) within the HDBSCAN cluster C_l .

Since the empirical return G_t is an unbiased estimator of the true state-action value, we have $\mathbb{E}[G_t | s_t, a_t] = Q^\pi(s_t, a_t)$. Assuming sufficient trajectory sampling within the cluster such that the baseline empirical average converges to the true expected state values within that region, i.e., $\mathbb{E}\left[\frac{1}{|C_l|} \sum_{s_i \in C_l} G_i\right] = \frac{1}{|C_l|} \sum_{s_i \in C_l} V^\pi(s_i)$, the expectation of the AGRPO advantage estimator is given by:

$$\mathbb{E}\left[\hat{A}^{\text{AGRPO}}(s_t, a_t)\right] = Q^\pi(s_t, a_t) - \frac{1}{|C_l|} \sum_{s_i \in C_l} V^\pi(s_i) \quad (7)$$

We analyze the absolute approximation error $\Delta(s_t, a_t) = \left| \mathbb{E}\left[\hat{A}^{\text{AGRPO}}(s_t, a_t)\right] - A^\pi(s_t, a_t) \right|$:

$$\Delta(s_t, a_t) = \left| \left(Q^\pi(s_t, a_t) - \frac{1}{|C_l|} \sum_{s_i \in C_l} V^\pi(s_i) \right) - (Q^\pi(s_t, a_t) - V^\pi(s_t)) \right| \quad (8)$$

$$= \left| V^\pi(s_t) - \frac{1}{|C_l|} \sum_{s_i \in C_l} V^\pi(s_i) \right| \quad (9)$$

Since $V^\pi(s_t)$ can be rewritten as $\frac{1}{|C_l|} \sum_{s_i \in C_l} V^\pi(s_i)$, we apply the triangle inequality:

$$\Delta(s_t, a_t) = \left| \frac{1}{|C_l|} \sum_{s_i \in C_l} (V^\pi(s_t) - V^\pi(s_i)) \right| \quad (10)$$

$$\leq \frac{1}{|C_l|} \sum_{s_i \in C_l} |V^\pi(s_t) - V^\pi(s_i)| \quad (11)$$

By Assumption 2, the value function is L_V -Lipschitz continuous. By Assumption 3, the maximum distance between any two states $s_t, s_i \in C_l$ is bounded by the cluster diameter ϵ_c . Therefore, $|V^\pi(s_t) - V^\pi(s_i)| \leq L_V \epsilon_c$ holds for all $s_i \in C_l$. Substituting this upper bound yields:

$$\Delta(s_t, a_t) \leq \frac{1}{|C_l|} \sum_{s_i \in C_l} L_V \epsilon_c = L_V \epsilon_c \quad (12)$$

This holds uniformly for any state $s_t \in C_l$, concluding the proof. $\square$

B.2 Proof of Theorem 4.2 (AGRPO Monotonic Policy Improvement)

Proof. To prove the monotonic improvement guarantee of AGRPO, we start with the exact performance difference lemma from Kakade and Langford [2002], which expresses the expected return of an updated policy $\tilde{\pi}$ in terms of the advantage function over the current policy π :

$$J(\tilde{\pi}) = J(\pi) + \frac{1}{1-\gamma} \mathbb{E}_{s \sim \rho_{\tilde{\pi}}, a \sim \tilde{\pi}} [A^\pi(s, a)] \quad (13)$$

where $\rho_{\tilde{\pi}}$ is the discounted state visitation distribution under $\tilde{\pi}$.

We substitute the true advantage $A^\pi(s, a)$ with our empirical AGRPO estimator. Let $\xi(s, a) = \mathbb{E}[\hat{A}^{\text{AGRPO}}(s, a)] - A^\pi(s, a)$ be the estimation error. From Lemma 4.1, we established that $-\xi(s, a) \geq -L_V \epsilon_c$. Consequently, $A^\pi(s, a) \geq \mathbb{E}[\hat{A}^{\text{AGRPO}}(s, a)] - L_V \epsilon_c$. Substituting this into Equation 13 yields:

$$J(\tilde{\pi}) \geq J(\pi) + \frac{1}{1-\gamma} \mathbb{E}_{s \sim \rho_{\tilde{\pi}}, a \sim \tilde{\pi}} [\hat{A}^{\text{AGRPO}}(s, a)] - \frac{L_V \epsilon_c}{1-\gamma} \quad (14)$$

Optimizing Equation 14 directly is intractable because the state distribution $\rho_{\tilde{\pi}}$ depends on the unknown updated policy $\tilde{\pi}$. Following Schulman et al. [2017], we approximate $\rho_{\tilde{\pi}}$ with the known state distribution ρ_π . The error introduced by this state distribution shift is bounded by the maximum total variation divergence $D_{\text{TV}}^{\max}(\pi, \tilde{\pi})$ between the policies.

Let $\epsilon_{\text{TV}} = \max_s D_{\text{TV}}(\pi(\cdot|s) \| \tilde{\pi}(\cdot|s))$. Theorem 1 of Schulman et al. [2017] establishes that the shift error is bounded by $\frac{2\gamma R_{\max}}{(1-\gamma)^2} \epsilon_{\text{TV}}^2$. Applying this bound to replace $\rho_{\tilde{\pi}}$ with ρ_π gives:

$$J(\tilde{\pi}) \geq J(\pi) + \frac{1}{1-\gamma} \mathbb{E}_{s \sim \rho_\pi, a \sim \tilde{\pi}} [\hat{A}^{\text{AGRPO}}(s, a)] - \frac{2\gamma R_{\max}}{(1-\gamma)^2} (D_{\text{TV}}^{\max}(\pi, \tilde{\pi}))^2 - \frac{L_V \epsilon_c}{1-\gamma} \quad (15)$$

In practice, AGRPO enforces the $D_{\text{TV}}^{\max}$ penalty intrinsically through the PPO clipping mechanism, which restricts the policy update ratio $r_t(\theta)$ within $[1 - \epsilon, 1 + \epsilon]$, ensuring the state distribution shift remains bounded. The cluster error term $\frac{L_V \epsilon_c}{1-\gamma}$ acts as an irreducible constant penalty determined by the HDBSCAN resolution ϵ_c . Therefore, as long as the expected localized advantage step outweighs the divergence penalty and the cluster radius error, the expected return of the updated policy strictly improves, concluding the proof. $\square$